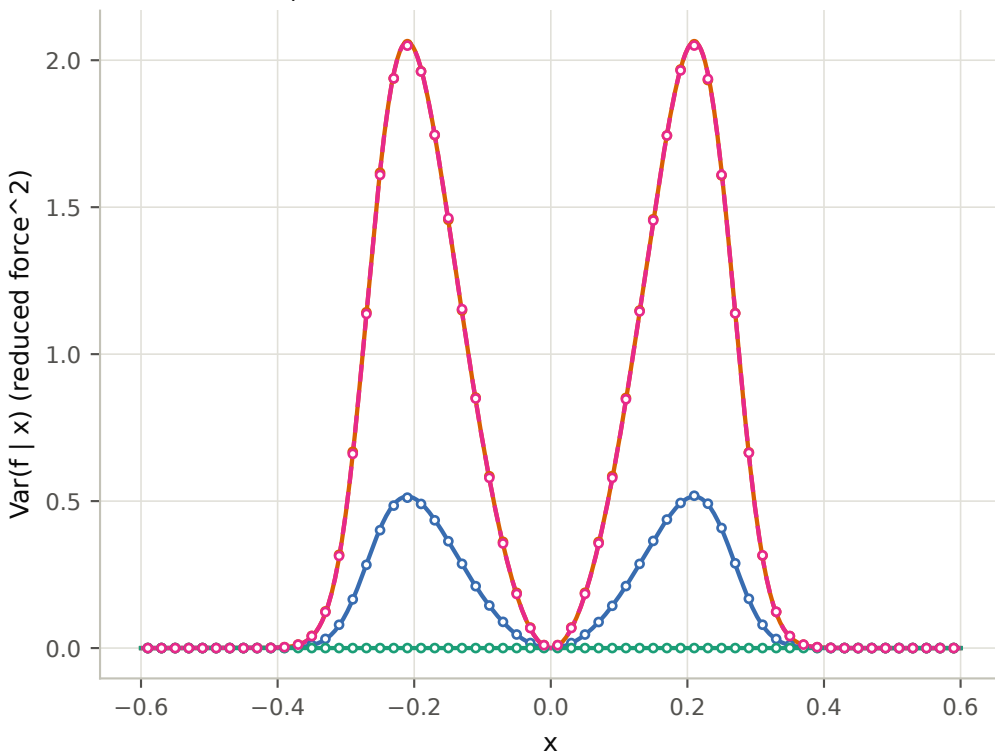


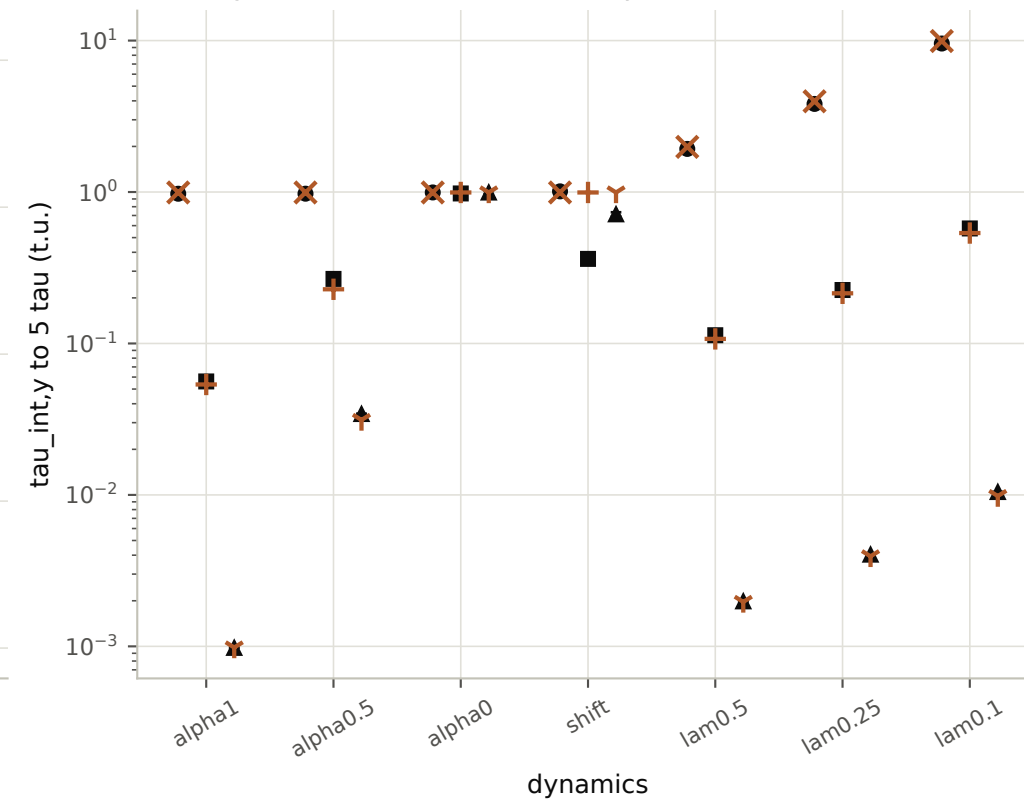
S4 conditional force variance and conditional relaxation per dynamics

validation summary present. Frozen-x OU: $\tau_y = 1/(\text{lam } k(x))$, $\tau_f = \tau_y / 2$ (alpha family) or τ_y (shift), integrated to 5 τ over the window $x_0 \pm 0.02$; no τ_f where f does not depend on y ($\alpha = 0$; $x = -1$, where $w' = 0$). alpha1, shift and the lambda dynamics share one $\text{Var}(f | x)$ (curves overlap).

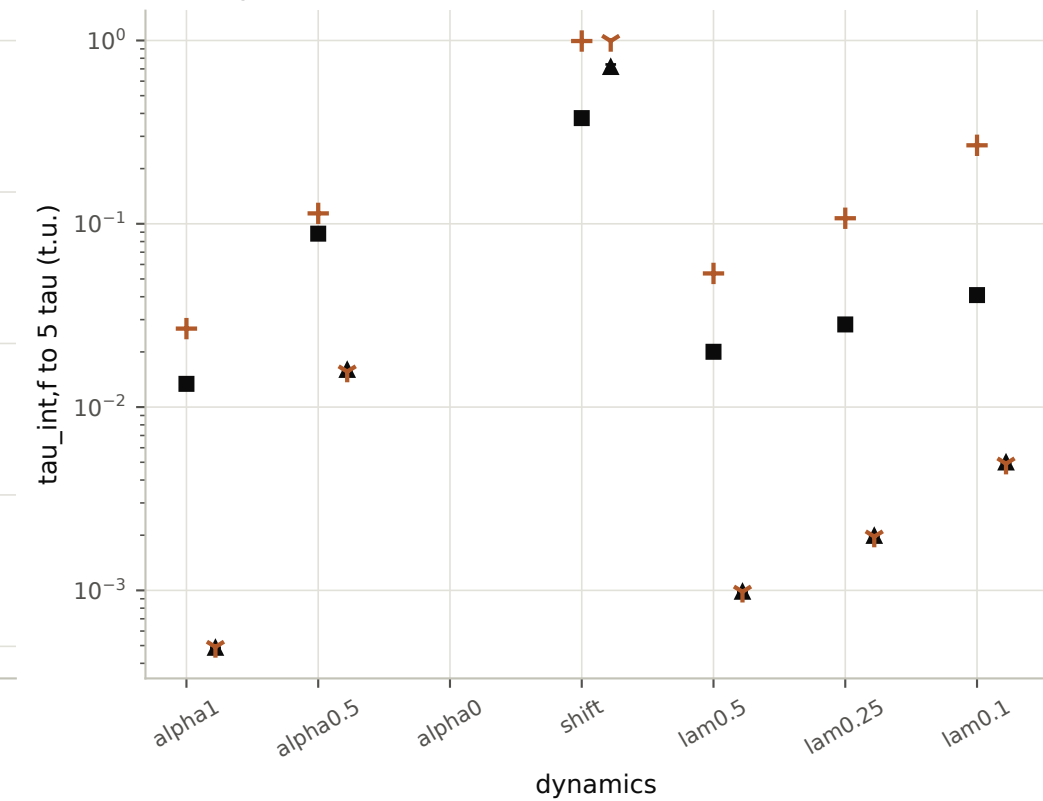
(a) exact $\text{Var}(f | x)$ and validation measurement



(b) integrated conditional ACF time of y



(c) integrated conditional ACF time of f



- (a) exact $\text{Var}(f|x)$, alpha1
- (a) exact $\text{Var}(f|x)$, alpha0.5
- (a) exact $\text{Var}(f|x)$, alpha0
- (a) measured $\text{Var}(f|x)$ (validation EM-flat chains, +2 se), in the model's colour
- × (b, c) frozen-x OU prediction, $x = -1$

- + (b, c) frozen-x OU prediction, $x = -0.21$
- Y (b, c) frozen-x OU prediction, $x = 0$
- (b, c) measured, $x = -1$ (+2 se)
- (b, c) measured, $x = -0.21$ (+2 se)
- ▲ (b, c) measured, $x = 0$ (+2 se)
- (b, c) measured, not resolved at the first lag